Optima Academy Online
AN EDUCATION EXPERIENCE COMPANY

Algebra 1 — Teacher Manual

FLORIDA 1200310 · GRADE 9 · 11 MODULES · FACILITATOR GUIDE · 2026–2027

Facilitator copy. Contains grading guidance, rubrics, and standards codes. **Do not publish to students.**

This manual documents the Algebra 1 course *as built* in Canvas — 11 modules delivered to roughly 500 asynchronous scholars, also offered as a live section. It is synthesized from the finished course. Anything that could not be verified from the build is flagged VERIFY for you to confirm against your Canvas instance before relying on it.

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1 · How this course works

Three concrete walk-throughs: a typical lesson, a typical week, and the major assessments.

A typical lesson

Every lesson follows the same shape, so a scholar always knows what to expect:

- **Lesson content page** (a Canvas Page) — the concept, worked examples, and Wiris-rendered equations. Read/worked, not graded.
- **Independent Practice** (a Canvas assignment built on Wiris quiz items) — about 4–10 problems, **auto-graded**, unlimited attempts, with per-question "check my answer" feedback and hints. Typically worth 10 points.
- **VR Lesson Resources page** — the immersive activity for that lesson (Activity to Complete / Location / IFXs / Tools). Reinforces the concept; produces no Canvas submission.
- Some lessons add a **Level-Up** page (a video or an escape-room style extension) for enrichment.

The graded touchpoint in a lesson is the Independent Practice, and it grades itself.

A typical week

Segment	What the scholar does	Graded?	Teacher time
Lesson content	Reads/works 1–2 lesson pages	No	~0
Independent Practice	4–10 problems per sub-lesson, retries to mastery	Auto	~0 (spot-check)
VR activity	Completes the lesson's VR resource	Completion	Light review
Mid-Module Check	1–2 per module; checks progress	Auto	~0
Study Hall	Optional live help + VR workshop	—	1 session

> **The key fact about this course's workload**
>
> Nearly every graded item is a Wiris auto-graded quiz. A facilitator's time goes to **outreach, Study Hall, and reviewing summative performance** — not to hand-grading. Plan your week around *watching the data and reaching out*, not a grading pile.

Major assessments

Assessment	Where	Grading	Notes
Independent Practice	Every sub-lesson	Auto, unlimited attempts	Formative; ~10 pts each
Mid-Module Check	1–2 per module	Auto	Progress check midway through a module
Summative Review	End of each module	Completion, unlimited attempts	Full-module review before the assessment
Summative Assessment	End of each module	Auto (graded for correctness), 2 attempts	~20 pts VERIFY point totals in your gradebook
Algebra 1 EOC	End of course (state)	State-scored	Cumulative; counts toward final grade per FL policy

> VERIFY **Honors items.** The build includes Honors-only variants (e.g., 2.04d, 3.04e, 9.02c, and Honors Mid-Module/Summative versions in Modules 9–11). In a base Algebra 1 section, publish/assign these only to Honors scholars; leave them unpublished otherwise.

2 · Adapting the async lessons to a live class that meets 2 days/week

Added at request. A repeatable weekly rhythm for a synchronous section that meets twice a week, using the same Canvas content unchanged.

The course is authored for self-paced study, so the live version does not replace it — it **wraps around it**. Scholars still read the lesson pages and complete the Independent Practices in Canvas; your two weekly meetings add the things async cannot: live worked examples, error analysis from real student data, partner practice, and the shared VR activity. The guiding split is "*content and practice happen in Canvas; sense-making and feedback happen live.*"

The standard two-day week

	Before class (async, in Canvas)	In the live meeting (~45–55 min)
Day 1 (new concept)	Read the week's first lesson content page(s); attempt its Independent Practice once (retries allowed).	**Warm-up** (3–5 min) on last week's sticking point → **mini-lesson** on the hardest idea of the week with one fully worked example → **guided practice** (you work one, they work one) → launch the **VR activity** or assign it as the between-class task.
Day 2 (practice & check)	Finish remaining Independent Practices and the VR activity for the week's lessons.	**Error analysis** from the Independent Practice data (pull the 2–3 most-missed items and work them) → **partner or station practice** on the mixed skills → scholars take the **Mid-Module Check** or **Summative Review** in class so you see it live → 2-minute preview of next week.

> **Module-end weeks**
>
> Use **Day 1** for a live Summative Review walkthrough (work the review together), then assign the **Summative Assessment** as the async task due before **Day 2**. Open Day 2 with targeted re-teaching of whatever the summative data shows scholars missed, then move into the next module.

> **VR in a live section**
>
> Because everyone is together, run the VR activity as a shared session on the campus when it fits Day 1, and use it as a gallery/debrief on Day 2. Scholars who cannot enter VR use the alternative path noted in the lesson so no one falls behind.

Suggested live pacing (2 meetings/week, ~36 weeks)

Proportional to the course's built day counts. Adjust to your calendar; Modules 3 and 8 span a quarter boundary.

Semester 1 (~16 wk)	Weeks	Semester 2 (~20 wk)	Weeks
M1 · Laws of Exponents & Radicals	3	M6 · Algebraic Reasoning	4
M2 · One-Variable Equations & Inequalities	4	M7 · Quadratic Equations & Functions	3
M3 · Linear Equations & Functions	5	M8 · Exponential Functions	3
M4 · Linear Inequalities & Abs. Value	2	M9 · Analyzing Functions	3
M5 · Systems of Linear Eq. & Ineq.	2	M10 · Comparing & Transforming Functions	3
		M11 · Statistical Data + EOC review	4

3 · Ungraded interactives & VR resources

- **Self-checking practice.** The Independent Practices, Mid-Module Checks, and Summative Reviews let a scholar check each answer as they go — right/wrong with a hint — and retry. This is formation-by-practice, not busywork. A scholar skipping the practice shows up indirectly in Summative performance, not as a missing grade.
- **VR Lesson Resources.** Each lesson has a VR activity page (Activity / Location / IFXs / Tools). It produces no submission; it reinforces the concept immersively. Review participation informally, not as a grade.
- **Level-Up pages.** Optional enrichment (videos, an "Escape Room: Radical Rescue"-style challenge). Purely for scholars who want to go further.

> VERIFY **No survey-mode intro/outro wrappers in this build.** Unlike some Optima courses, this Algebra 1 export does not use Canvas survey-quiz module wrappers that capture essential-question or confidence data. Your best early-warning signal here is the **Mid-Module Check and Summative auto-grade data** plus module-progress completion — use those for triage (see §11).

4 · AI & VR in this course (student-facing)

AI. The syllabus states the posture: scholars may use AI to support learning (explain a concept, check understanding, see a worked example) unless an assignment says otherwise, but AI may never do the thinking or the work — the problem-solving and the steps must be their own, and use should be disclosed. Speak to this positively: in math the point is to be able to judge whether an answer is right, which requires doing the work yourself.

VR. Every lesson has an immersive activity on Optima's VR campus that reinforces the concept and produces no submission. Scholars keep a headset charged; those who cannot enter VR use the alternative path in the lesson. Describe VR by what scholars do, not by a platform name.

5 · Using Ignite AI to support students (teacher-facing)

> VERIFY Ignite AI feature names and availability vary by Canvas release and by what Optima has enabled. Confirm the current feature set against your Canvas instance; describe capabilities functionally rather than promising a specific button.

Use the Canvas AI assistant to make a 500:1 ratio feel personal — always as a **draft the teacher edits and owns**, never to surveil or auto-penalize:

- **Summarize summative/Mid-Module analytics** to find the cohort's most-missed skills fast, then plan a targeted announcement or Study Hall focus.
- **Surface scholars who need a nudge** (stalled module progress, repeated low summative attempts) and draft a warm check-in you personalize before sending.
- **Generate practice variants** for a scholar who needs another rep on a Tier-1 skill (e.g., more factoring problems like 6.02).
- **Communicate with families** in clear, kind summaries; translate when helpful.

Standing rule: AI drafts, the teacher decides. You read, edit, and own anything sent to a scholar or family.

6 · Grading categories & Canvas gradebook setup

A click-level setup so a new facilitator configures the gradebook consistently. Confirm official weights with your school.

Assignment group	What's in it	Suggested weight	Grading method
Practice & Checks	Independent Practices, Mid-Module Checks	40%	Auto (Wiris)
Summative Assessments	Module Summative Assessments	40%	Auto, 2 attempts
VR & Participation	VR activities, Summative Reviews (completion)	20%	Completion

1. **Gradebook → Settings → enable "Weight final grade based on assignment groups."**
2. Create the three groups above and set the weights (confirm with your school; the FL **EOC** is applied to the final course grade separately per policy).
3. Mark Summative Reviews and VR items as **Complete/Incomplete**, not points.
4. Set your late/missing policy to match the syllabus: no per-date penalty, but end-of-quarter zero for missing work and a 10% deduction in the final 14 days of a quarter.
5. Use the **module requirements / progress view** as the at-a-glance completion signal across the ungraded scaffolding.

> VERIFY Point values above (10-pt practices, 20-pt summatives) are read from the build's assignment settings and should be confirmed in your gradebook after import.

7 · Rubrics for teacher-graded items

This course is **predominantly auto-graded** (Wiris). Most modules need no rubric. Use the rubric below only where a scholar submits work you score by hand — primarily the **Lesson 5.03a Systems of Constraints** project (which carries PROJECT / PROBLEMS / SUBMISSION sections) and any free-response a summative may contain.

Standard math task rubric (4 criteria — keep identical across items)

Criterion	Full credit	Partial	Minimal
Mathematical accuracy	Correct answer and correct method throughout	Right approach, minor computational errors	Method or answer largely incorrect
Reasoning & process	Steps shown clearly and justified	Some steps shown; gaps in justification	Little or no work shown
Modeling / representation	Correctly sets up the model, graph, or system for the situation	Partial or slightly mismatched setup	Setup missing or unrelated to the task
Clarity & precision	Labels, units, and notation correct and readable	Some labels/units missing	Unclear or unlabeled

How to attach it

Paste this as a native **Canvas Rubric** on the item so it scores in SpeedGrader. Keep the four criteria identical across any hand-graded item so scoring stays fast and consistent.

VERIFY Whether any Module Summative contains a free-response question was not confirmable from the build (items render as Wiris quiz objects). Check each Summative in Canvas; if a free-response exists, score it with the rubric above and write a 2–3 point answer guide for it here.

8 · Assessment & interactive catalog

Type	What it is	How it's scored	Recommended teacher move
Independent Practice	Per-sub-lesson Wiris problem set	Auto, unlimited attempts, per-question check + hint	Don't grade. Scan the item analysis for most-missed skills.
Mid-Module Check	Midway progress quiz	Auto	Use the data to decide what to re-teach or announce.
Summative Review	Full-module review	Completion, unlimited attempts	Encourage before the assessment; use live in a 2-day section.
Summative Assessment	End-of-module graded quiz	Auto, 2 attempts	Watch score distribution; reach out to low/repeated-attempt scholars.
VR Lesson Resources	Immersive per-lesson activity	Completion / informal	Debrief in Study Hall; assemble a gallery in a live section.
Level-Up	Optional enrichment (video/escape room)	Ungraded	Offer to scholars ready to go further.

9 · Coach, not catch

- Feedback names the next step, not just the error. Lead with what's working.
- Unlimited attempts on practice are a learning tool, not a trap — encourage retries to mastery.
- Use analytics and Ignite AI to **find scholars who need help early**, then reach out — not to build a case.
- On suspected AI misuse or copied work: open a conversation, ask the scholar to talk through their steps, treat it as formation.

10 · Weekly announcement scripts

Asynchronous scholars rely on your weekly announcement as your main presence. Clone the template; the drafts below match this course's modules.

TEMPLATE — [MODULE N / WEEK]

This week you should be in **[module · lessons]**. Focus on **[the one key skill]**. Don't miss **[the due item, e.g., the Module N Summative by [date]]**. One tip: **[a common pitfall]**. Stuck? Bring it to Study Hall on **[day/time]** — that's what it's for. You've got this.

MODULE 1 — LAWS OF EXPONENTS & RADICALS

Welcome to Algebra 1! This week we simplify radicals and apply the laws of exponents. The classic trap is losing track of signs and the rules for rational exponents — write each rule down before you practice. Finish the 1.01 and 1.02 Independent Practices, and try the VR activity. Study Hall is [day/time].

MODULE 6 — ALGEBRAIC REASONING (heavy)

This is a big module — polynomials, factoring, and solving quadratics three ways. Pace yourself across the two weeks and don't skip the factoring practice (6.02); everything after it depends on factoring fluency. The Summative has two attempts, so use the Summative Review first.

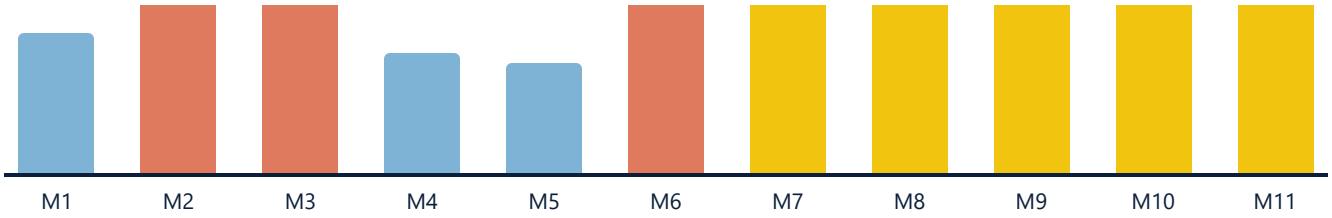
SITUATIONAL — PRE-SUMMATIVE NUDGE

Module [N] Summative opens this week. Do the Summative Review first, check every answer, then take the assessment — you have two attempts. If your Mid-Module Check felt shaky on [skill], come to Study Hall before you start.

11 · Teacher workflow & workload map

A normal week, in order: (1) review Mid-Module/Summative analytics for the modules in play, (2) message the scholars the data flags, (3) post the weekly announcement, (4) host Study Hall + the VR workshop, (5) answer messages. Because grading is automated, **outreach and Study Hall are the job** — budget your time there.





Relative facilitator *support* load by module (outreach, VR review, summative oversight) — scaled to module size, since hand-grading is near zero. Blue = light, gold = moderate, coral = heavy. Plan extra outreach time on the coral modules (M2, M3, M6). M3 and M8 each span a quarter boundary.

12 · Standards coverage map (Florida B.E.S.T. → modules)

Added at request. Where each Florida B.E.S.T. standard is taught, by module. Codes are teacher-copy only — never echoed to students. Honors-only standards are marked **(H)**. Read from the course's Scope & Sequence.

Module	Standards covered
M1 · Laws of Exponents & Radicals	NSO.1.1 (exponents, numerical) · NSO.1.2 (exponents, algebraic) · NSO.1.4 (radicals & operations)
M2 · One-Variable Equations & Inequalities	AR.2.1 (linear equations) · AR.2.6 (linear inequalities) · AR.4.1 (absolute value equations) · AR.1.2 (rearranging formulas) · FL.3.2 (simple interest) · **(H) AR.4.2 (absolute value inequalities)**
M3 · Linear Equations & Functions	AR.2.4 (graphing linear) · AR.2.2 (writing linear equations) · AR.2.5 (modeling with linear) · AR.2.3 (parallel & perpendicular) · DP.2.4 (lines of fit) · DP.2.6 (using residuals) · **(H) DP.2.5 (plot & analyze residuals)**
M4 · Linear Inequalities & Absolute Value	AR.2.8 (graphing 2-var inequalities) · AR.2.7 (writing 2-var inequalities) · AR.4.3 (graphing 2-var absolute value)
M5 · Systems of Linear Eq. & Ineq.	AR.9.1 (systems of equations) · AR.9.4 (systems of inequalities) · AR.9.6 (constraints as systems)
M6 · Algebraic Reasoning	AR.1.1 (parts of an expression) · AR.1.3 (polynomial operations) · AR.1.4 (dividing by a monomial) · AR.1.7 (rewriting/factoring polynomials) · AR.3.1 (one-variable quadratic equations)
M7 · Quadratic Equations & Functions	AR.3.6 (vertex & zeros) · AR.3.7 (graphing quadratics) · AR.3.4 (writing quadratic equations) · AR.3.5 (from x-intercepts & a point) · AR.3.8 (modeling with quadratics)
M8 · Exponential Functions	AR.5.3 (classifying exponentials) · AR.5.6 (graphing exponentials) · AR.5.4 (writing exponentials) · FL.3.2 (compound interest) · FL.3.4 (simple vs compound → linear vs exponential)
M9 · Analyzing Functions	F.1.1 (classifying function types) · F.1.8 (best model) · F.1.2 (evaluating functions) · F.1.3 (average rate of change) · **(H) F.3.1 (combining functions)**
M10 · Comparing & Transforming Functions	F.1.5 (comparing linear) · F.1.6 (linear vs non-linear) · F.2.1 (effects of transformations) · **(H) F.2.3 (state transformation & find k)**
M11 · Statistical Data	DP.1.1 (categorical/numerical, uni/bivariate) · DP.1.2 (interpreting distributions) · DP.1.3 (correlation vs causation) · DP.1.4 (surveys & margin of error) · DP.3.1 (two-way tables) · **(H) DP.3.2 (tables from relative frequencies)** · **(H) DP.3.3 (joint/marginal/conditional frequencies)**

Spanning standards to watch

FL.3.2 (simple & compound interest) is split: simple interest in **M2**, compound in **M8**. Modules **3** and **8** each span a quarter boundary at their Mid-Module Check, so their standards carry across two quarters.

13 · What makes this course classical & cultivating culture

Classical in a math course shows up as *wonder before mechanism* and *reasoning made visible*: scholars explore a pattern (parallel/perpendicular slopes, the shape of growth vs. decay) before they are handed the rule, and they justify their steps rather than only producing answers. The commonplace practice extends to mathematics — scholars keep the ideas and definitions worth remembering.

Cultivating culture asynchronously: name scholars' good reasoning publicly in announcements; model curiosity ("here's a slick second way to solve 6.03c"); hold Study Hall as a real gathering, not just office hours; and treat feedback as relationship. In a live section, the two weekly meetings are where the community forms — protect the warm-up and debrief.

14 · Sensitive-content notes

Algebra 1 carries **no mature-theme or contested content**. The only advance notes worth giving families are practical: this is a **VR course** (headset required; an alternative path exists for any scholar who cannot use VR), and it is a **state-tested course** ending in the Algebra 1 EOC. Both are covered in the syllabus; no special content script is needed.